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1. The infinite series C and S are defined by

$$C = \cos 2\theta - \frac{1}{3} \cos 5\theta + \frac{1}{9} \cos 8\theta - \frac{1}{27} \cos 11\theta + \dots$$
$$S = \sin 2\theta - \frac{1}{3} \sin 5\theta + \frac{1}{9} \sin 8\theta - \frac{1}{27} \sin 11\theta + \dots$$

Given that the series C and S are both convergent.

(a) Show that

$$C + iS = \frac{3e^{2i\theta}}{3 + e^{3i\theta}} \quad [4]$$

(b) Hence show that

$$S = \frac{9 \sin 2\theta - 3 \sin \theta}{10 + 6 \cos 3\theta} \quad [4]$$

2. (i) Show that

$$(1 - 2e^{2i\theta})(1 - 2e^{-2i\theta}) = 5 - 4\cos 2\theta \quad [3]$$

For a positive integer n , series C and S are defined by

$$C = \cos \theta + 2 \cos 3\theta + 4 \cos 5\theta + \cdots + 2^{n-1} \cos((2n-1)\theta)$$

$$S = \sin \theta + 2 \sin 3\theta + 4 \sin 5\theta + \cdots + 2^{n-1} \sin((2n-1)\theta)$$

(ii) Show that

$$C = \frac{2^{n+1} \cos((2n-1)\theta) - 2^n \cos((2n+1)\theta) - \cos \theta}{5 - 4 \cos 2\theta}$$

and find a similar expression for S .

[9]

3. (i) Show that

$$1 - e^{i2\theta} = 2 \sin \theta (\sin \theta - i \cos \theta) \quad [3]$$

(ii) For a positive integer n , the series C and S are defined as follows.

$$\begin{aligned} C &= 1 - \binom{n}{1} \cos 2\theta + \binom{n}{2} \cos 4\theta - \dots + (-1)^n \cos 2n\theta \\ S &= -\binom{n}{1} \sin 2\theta + \binom{n}{2} \sin 4\theta - \dots + (-1)^n \sin 2n\theta \end{aligned}$$

By considering $C + iS$, show that

$$C = 2^n \sin^n \theta \cos \left(n\theta - \frac{n\pi}{2} \right)$$

and find a corresponding expression for S .

[9]

4. The infinite series C and S are defined as follows:

$$C = \cos \alpha + r \cos(\alpha + \beta) + r^2 \cos(\alpha + 2\beta) + \dots$$

$$S = \sin \alpha + r \sin(\alpha + \beta) + r^2 \sin(\alpha + 2\beta) + \dots$$

where r is a real number and $|r| < 1$.

By considering $C + iS$, show that

$$C = \frac{\cos \alpha - r \cos(\alpha - \beta)}{1 + r^2 - 2r \cos \beta}$$

and find a corresponding expression for S .

[8]

5. (i) For values of θ for which the expressions are defined, show that

$$1 + i \tan \theta = \sec \theta (\cos \theta + i \sin \theta) \quad [2]$$

- (ii) For a positive integer n , the series C and S are defined by

$$C = 1 - \binom{n}{2} \tan^2 \theta + \binom{n}{4} \tan^4 \theta - \dots$$

$$S = \binom{n}{1} \tan \theta - \binom{n}{3} \tan^3 \theta + \binom{n}{5} \tan^5 \theta - \dots$$

where in each case the pattern continues until the power of $\tan \theta$ is at most n .

By considering $C + iS$, show that

$$C = \sec^n \theta \cos n\theta$$

and find a corresponding expression for S .

[7]

6. The convergent series C and S are defined by

$$C = \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n \cos((4n+1)\theta)$$

$$S = \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n \sin((4n+1)\theta)$$

(a) Show that

$$C + iS = \frac{2e^{i\theta}}{2 + e^{4i\theta}} \quad [4]$$

(b) Hence show that

$$C = \frac{4 \cos \theta + 2 \cos 3\theta}{5 + 4 \cos 4\theta} \quad [4]$$

7. Let

$$C = \sum_{r=0}^{n-1} \cos^2 \left(\theta + \frac{2\pi r}{n} \right)$$
$$S = \sum_{r=0}^{n-1} \sin \left(\theta + \frac{2\pi r}{n} \right) \cos \left(\theta + \frac{2\pi r}{n} \right)$$

where n is an integer greater than 2. Let $\exp(x)$ denote e^x .

By considering the real and imaginary parts of

$$\sum_{r=0}^{n-1} \exp \left(2i \left(\theta + \frac{2\pi r}{n} \right) \right)$$

show that $C = \frac{n}{2}$ and $S = 0$.

[7]